

Semidefinite Programming in Quantum Information
Lecture 2: SDP Relaxations, PPT, and the Diamond Norm
Quantum in the Dunes

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What we will do today

- 1 Why SDPs as relaxations in QIT?
- 2 Bipartite entanglement and the PPT relaxation
 - Separable, entangled, PPT
 - Optimizing over PPT states is an SDP
 - Example: maximally entangled overlap
- 3 Linear maps: Stinespring, Choi, and norms
 - Stinespring-style representations
 - Norms and operational tasks
- 4 The diamond norm as an SDP
 - Operational meaning
 - Maximum output fidelity route
 - Fidelity as an SDP
- 5 Complexity signposts: nearby norms can be hard
 - Induced norms and contraction coefficients
- 6 Wrap-up

Where SDPs enter: two recurring patterns

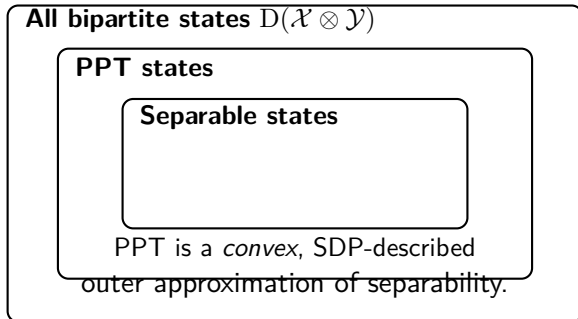
Pattern A: hard physics set $\rightarrow$ SDP-friendly outer approximation

- Separable states, LOCC measurements, quantum correlations, ... are complicated sets.
- Replace by *tractable* convex supersets (PPT, k -extendible, NPA levels, ...).
- Solve the relaxed SDP to get bounds / certificates.

Pattern B: dual variables are witnesses

- Primal feasible point $\Rightarrow$ a *strategy* (constructive lower bound).
- Dual feasible point $\Rightarrow$ a *certificate* (upper bound / witness).

A picture to keep in mind



Separable vs. entangled states

Let $\rho_{AB} \in \mathcal{D}(\mathcal{X} \otimes \mathcal{Y})$.

Definition (separability)

ρ_{AB} is **separable** if it is a convex combination of product states:

$$\rho_{AB} = \sum_i p_i \rho_i^A \otimes \rho_i^B \quad (p_i \geq 0, \sum_i p_i = 1).$$

Otherwise it is **entangled**.

- The set of separable states is convex but has a complicated boundary.
- Optimizing a linear functional over separable states is generally hard.

Partial transpose and the PPT cone

Fix product bases on A and B . The **partial transpose** on B is

$$(\text{id} \otimes T)(X) = \sum_{i,j} (\text{id} \otimes T)(X_{ij} \otimes |i\rangle\langle j|) = \sum_{i,j} X_{ij} \otimes |j\rangle\langle i|.$$

Definition (PPT)

A bipartite state ρ_{AB} is **PPT** if $\rho_{AB}^{T_B} := (\text{id} \otimes T)(\rho_{AB}) \succeq 0$.

Key fact (Peres)

If ρ_{AB} is separable, then ρ_{AB} is PPT.

Why PPT is attractive

- The constraint $\rho_{AB} \succeq 0$ is already SDP-friendly.
- Adding $\rho_{AB}^{T_B} \succeq 0$ preserves convexity and yields another PSD constraint.
- PPT is a *physically meaningful* relaxation: it detects many entangled states (NPT states).

Dimension caveat

For 2×2 and 2×3 systems, $\text{PPT} \Leftrightarrow \text{separable}$. In larger dimensions PPT is only a relaxation.

A template: optimize over PPT states

Let $H \in \text{Herm}(\mathcal{X} \otimes \mathcal{Y})$ be a Hermitian *test operator* (energy, witness, payoff).

PPT-relaxed optimization

$$\text{maximize: } \text{Tr}(H\rho_{AB})$$

$$\text{subject to: } \text{Tr}(\rho_{AB}) = 1, \quad \rho_{AB} \succeq 0, \quad \rho_{AB}^{T_B} \succeq 0.$$

- Same objective as the “true” separable optimization, but larger feasible set.
- The optimal value is an *upper bound* on the separable optimum.

Dual viewpoint: PPT witnesses

The dual of a PPT optimization typically has the form

$$\text{find } (Y, Z, \lambda) \text{ s.t. } \lambda \mathbb{1} - H = Y + (\text{id} \otimes T)(Z), \quad Y, Z \succeq 0,$$

with objective $\min \lambda$ (up to conventions).

Interpretation

- $Y \succeq 0$ and $(\text{id} \otimes T)(Z)$ are the “cones” supporting the relaxation.
- Dual feasible points certify: for all PPT states ρ , $\text{Tr}(H\rho) \leq \lambda$.

Example: maximum overlap with a maximally entangled state

Let $\dim(\mathcal{X}) = \dim(\mathcal{Y}) = d$ and

$$|\Phi\rangle = \frac{1}{\sqrt{d}} \sum_{i=1}^d |i\rangle |i\rangle, \quad \Phi := |\Phi\rangle\langle\Phi|.$$

Consider

$$\max\{\mathrm{Tr}(\Phi \rho_{AB}) : \rho_{AB} \in \mathcal{D}(\mathcal{X} \otimes \mathcal{Y}), \rho_{AB}^{T_B} \succeq 0\}.$$

Fact

Every PPT state has overlap at most $1/d$ with Φ (and the bound is tight).

Proof idea: one-line dual certificate

- Note that $\Phi^{T_B} = \frac{1}{d} \text{SWAP}$ and $\text{SWAP} \preceq \mathbb{1}$.
- If $\rho^{T_B} \succeq 0$, then

$$\text{Tr}(\Phi\rho) = \text{Tr}(\Phi^{T_B}\rho^{T_B}) \leq \text{Tr}\left(\frac{1}{d} \mathbb{1} \cdot \rho^{T_B}\right) = \frac{1}{d}.$$

Takeaway

PPT constraints can yield *clean analytic bounds* via simple dual-feasible inequalities.

Stinespring: channels as isometries + environment

A CPTP map $\Phi : \mathcal{L}(\mathcal{X}) \rightarrow \mathcal{L}(\mathcal{Y})$ admits a Stinespring dilation:

$$\Phi(X) = \text{Tr}_{\mathcal{Z}}(V X V^*) \quad \text{for some isometry } V : \mathcal{X} \rightarrow \mathcal{Y} \otimes \mathcal{Z}.$$

Equivalently (Kraus form), $\Phi(X) = \sum_i K_i X K_i^*$ with $\sum_i K_i^* K_i = \mathbb{1}$.

Why this matters

Stinespring makes the *geometry* of complete positivity explicit: positivity is enforced by “sandwiching” X between operators.

A Stinespring-type representation for *general* linear maps

Even if Φ is not CP, one can write it using *two* operators:

$$\Phi(X) = \text{Tr}_{\mathcal{Z}}(A_0 X A_1^*), \quad A_0, A_1 \in \text{L}(\mathcal{X}, \mathcal{Y} \otimes \mathcal{Z}).$$

Why this matters here

This representation is the starting point for the diamond norm SDP via *maximum output fidelity*.

Norms for maps: induced trace norm is not stable

For $\Phi : L(\mathcal{X}) \rightarrow L(\mathcal{Y})$ define

$$\|\Phi\|_{1 \rightarrow 1} := \max\{ \|\Phi(X)\|_1 : \|X\|_1 \leq 1 \}.$$

Problem

Tensoring with an identity can increase this norm:

$$\|\Phi \otimes \text{id}\|_{1 \rightarrow 1} > \|\Phi\|_{1 \rightarrow 1} \quad \text{can happen.}$$

In QIT, allowing entanglement with an ancilla is operationally natural, so we need a *stabilized* norm.

Diamond norm / completely bounded trace norm

Definition

$$||| \Phi |||_1 := \sup_{k \geq 1} \|\Phi \otimes \text{id}_{\mathbb{C}^k}\|_{1 \rightarrow 1}.$$

When Φ is Hermiticity-preserving, this is the **diamond norm** $\|\Phi\|_{\diamond}$.

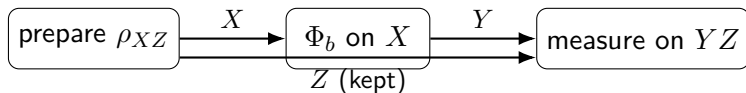
Finite stabilization

It suffices to take $k = \dim(\mathcal{X})$.

- $\|\cdot\|_{\diamond}$ is often the “right” distance between channels in operational settings.
- Crucially: it admits an efficient SDP.

Channel discrimination: the operational meaning

We are promised that the unknown channel is Φ_0 or Φ_1 , each with prior $1/2$.



Channel Holevo–Helström theorem

$$p_{\text{succ}}^* = \frac{1}{2} + \frac{1}{4} \|\Phi_0 - \Phi_1\|_{\diamond}.$$

From Stinespring pairs to maximum output fidelity

Given $\Phi(X) = \text{Tr}_{\mathcal{Z}}(A_0 X A_1^*)$ define CP maps $\Psi_0, \Psi_1 : L(\mathcal{X}) \rightarrow L(\mathcal{Z})$ by

$$\Psi_0(X) = \text{Tr}_{\mathcal{Y}}(A_0 X A_0^*), \quad \Psi_1(X) = \text{Tr}_{\mathcal{Y}}(A_1 X A_1^*).$$

Theorem (Watrous)

$$\|\Phi\|_1 = \max \left\{ F(\Psi_0(\rho_0), \Psi_1(\rho_1)) : \rho_0, \rho_1 \in \mathcal{D}(\mathcal{X}) \right\}.$$

Thus: diamond norm $\leftrightarrow$ an optimization of *fidelity* between two CP outputs.

Fidelity: definition and a key SDP characterization

For PSD P, Q the Uhlmann fidelity is

$$F(P, Q) = \|\sqrt{P}\sqrt{Q}\|_1^2.$$

Useful equivalent formulation

$$\sqrt{F(P, Q)} = \max \left\{ \frac{1}{2} \operatorname{Tr}(Y + Y^*) : \begin{pmatrix} P & Y \\ Y^* & Q \end{pmatrix} \succeq 0 \right\}.$$

- The block PSD constraint is convex in (P, Q, Y) .
- This turns a nonlinear quantity (a trace norm of a product) into an SDP.

Fidelity: SDP characterization via Schur complements

Proof sketch (Schur complement $\Rightarrow$ contraction factorization)

Assume $Q \succ 0$ (otherwise replace Q by $Q + \varepsilon \mathbb{1}$ and let $\varepsilon \downarrow 0$ at the end). Then by the Schur complement,

$$\begin{pmatrix} P & Y \\ Y^* & Q \end{pmatrix} \succeq 0 \iff P - YQ^{-1}Y^* \succeq 0.$$

Let $W := YQ^{-1/2}$. The condition becomes $WW^* \preceq P$, which is equivalent to the existence of a contraction K with $\|K\|_\infty \leq 1$ such that

$$W = P^{1/2}K \iff Y = P^{1/2}KQ^{1/2}.$$

Hence the objective is

$$\frac{1}{2} \operatorname{Tr}(Y + Y^*) = \Re \operatorname{Tr}(K Q^{1/2} P^{1/2}) \leq \|K\|_\infty \|Q^{1/2} P^{1/2}\|_1 \leq \|\sqrt{P} \sqrt{Q}\|_1,$$

using Hölder duality $\sup_{\|K\|_\infty \leq 1} \Re \operatorname{Tr}(KA) = \|A\|_1$. Taking the polar decomposition $Q^{1/2} P^{1/2} = U |Q^{1/2} P^{1/2}|$ and choosing $K = U^*$ attains equality, giving $\max = \|\sqrt{P} \sqrt{Q}\|_1 = \sqrt{F(P, Q)}$.

Primal SDP: maximum output fidelity

Decision variables: $\rho_0, \rho_1 \in \mathcal{D}(\mathcal{X})$ and $Y \in \mathcal{L}(\mathcal{Z})$.

SDP

$$\begin{aligned} & \text{maximize: } \frac{1}{2} \text{Tr}(Y + Y^*) \\ & \text{subject to: } \begin{pmatrix} \Psi_0(\rho_0) & Y \\ Y^* & \Psi_1(\rho_1) \end{pmatrix} \succeq 0, \\ & \quad \rho_0 \succeq 0, \text{Tr}(\rho_0) = 1, \quad \rho_1 \succeq 0, \text{Tr}(\rho_1) = 1. \end{aligned}$$

Output

The optimum equals $\|\Phi\|_1$ (hence $\|\Phi\|_\diamond$ for Hermiticity-preserving Φ).

How you actually use primal vs. dual

Primal feasible $\Rightarrow$ lower bound

- Choose (ρ_0, ρ_1, Y) feasible.
- Interpretable as an explicit discrimination strategy.
- Certifies: $\|\Phi\|_{\diamond} \geq \frac{1}{2}\text{Tr}(Y + Y^*)$.

Dual feasible $\Rightarrow$ upper bound

- Choose $Z \succ 0$ and compute the objective.
- Interpretable as a witness/certificate.
- Certifies: no strategy beats this bound.

Induced Schatten norms (unstabilized)

For $1 \leq q, p \leq \infty$ define

$$\|\Phi\|_{q \rightarrow p} = \max\{ \|\Phi(X)\|_p : \|X\|_q \leq 1 \}, \quad \|\Phi\|_{q \rightarrow p}^+ = \max\{ \|\Phi(X)\|_p : X \succeq 0, \|X\|_q \leq 1 \}.$$

Caution

Even when Φ is completely positive (or a difference of channels), these *unstabilized* norms can encode hard combinatorics.

Hardness without complete boundedness

Recent results (see the attached preprints) show NP-completeness / NP-hardness phenomena such as:

- Computing $\|\Phi\|_{1 \rightarrow 1}^+$ can be NP-complete for maps presented as differences of channels.
- For every fixed $p > 1$, computing $\|\Phi\|_{1 \rightarrow p}$ can be NP-complete even for completely positive, entanglement-breaking maps.
- Approximating certain contraction (ergodicity) coefficients within constant factors can be NP-hard.

Moral

“Looks like a norm” does not mean “has an SDP.”

CB norms remain SDP-friendly (often)

Define the completely bounded $1 \rightarrow p$ norms

$$\|\Phi\|_{\text{cb}, 1 \rightarrow p} := \sup_{k \geq 1} \|\Phi \otimes \text{id}_k\|_{1 \rightarrow p}.$$

Contrast with unstabilized norms

For fixed p , these CB norms retain SDP descriptions (diamond norm is $p = 1$).

- Complete boundedness brings additional operator-system structure.
- In some regimes (e.g. $q \geq p$ for CP maps) even $\|\Phi\|_{q \rightarrow p}$ can reduce to eigenvalue-type optimizations.

Contraction coefficients

For trace distance:

$$\eta_{\text{tr}}(\Phi) := \sup_{\rho \neq \sigma} \frac{\|\Phi(\rho - \sigma)\|_1}{\|\rho - \sigma\|_1}.$$

- Controls mixing / distinguishability decay under a channel.
- But can be computationally hard to approximate, and some decision variants are NP-hard.

Useful mental model

If the definition optimizes over *product* inputs, be cautious about hardness.

If it optimizes over *arbitrarily entangled* inputs (CB norm), an SDP is more likely.

Takeaways

- **PPT is the first workhorse relaxation** for bipartite entanglement: convex, SDP-described, and often analytically tractable via dual certificates.
- **Diamond norm is operational and SDP-computable:** channel discrimination reduces to a maximum output fidelity SDP via Stinespring pairs.
- **Complexity is delicate:** small changes to stabilization or exponents can flip tractable $\leftrightarrow$ intractable.

Further reading

- J. Watrous, *Theory of Quantum Information* (lecture notes / book draft).
- Watrous: SDPs for the completely bounded trace norm (arXiv:1207.5726).
- Complexity of induced norms / contraction coefficients: see attached preprints (arXiv:2507.08358, arXiv:2507.16737).